

$$G(s) = \frac{100}{s(10s+1)(s+100)}$$

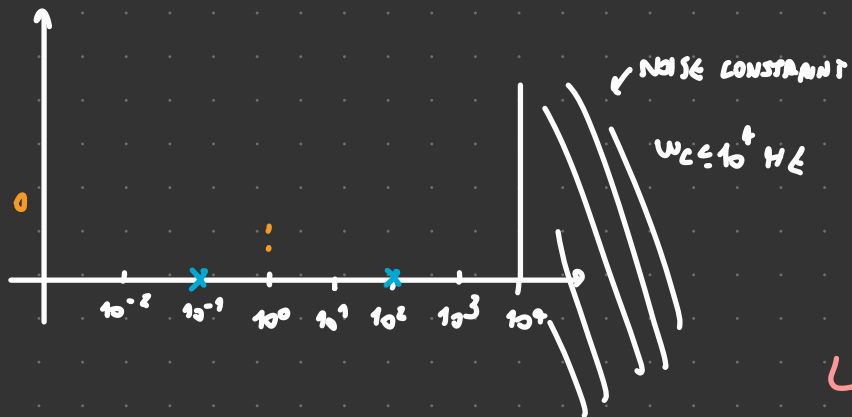
- REAL
- $\zeta \geq 15\%$
- $\sigma \leq 5\%$
- $T_s \leq 10 \text{ SEC}$
- NOISE ATTENUATION AT LEAST 20 dB

• ERROR SATISFIED SINCE $q > 1$

• $\zeta \geq 0.15 \rightarrow \varphi_m \geq 15^\circ$

$$T_s \leq 10 \rightarrow 2.2T \leq 10 \rightarrow T \leq 4.5 \text{ SECONDS} \rightarrow \frac{1}{\omega_c} \leq 4.5 \text{ SECONDS} \rightarrow \omega_c \geq 1/4.5 \rightarrow \omega_c \geq 0.22 \text{ Hz}$$

↓
I CHOOSE $\omega_c = 10^0 \text{ Hz}$



$$\omega_c = 10^0$$

$$s = -10^0 j$$

$$L = \frac{100}{s(10s+1)(s+100)}$$

$$\tan^{-1}(0/100) - \tan^{-1}(10^0/0)$$

$$- \tan^{-1}(10 \cdot 10^0 / 1)$$

$$- \tan^{-1}(10^0 / 100)$$